

Skew lines

The case the plane does not have — its test, its angle, and the common perpendicular that measures the distance between two lines that never meet.

LESSON 21–22

2 × 40 MIN

GEOMETRY 10, §10

EXTENSION

LESSON CLOCK

10'

22'

24'

20'

20'

4

The test	10 min
The angle, by translation	22 min
The common perpendicular	24 min
Distances in a cuboid	20 min
Practice	20 min
Homework	4 min

LEARNING OBJECTIVES

- State and apply the test for skew lines.
- Find the angle between two skew lines by translation.
- Describe the common perpendicular of two skew lines.
- Compute the distance between skew edges of a cuboid.

TEXTBOOK

National	pp. 93–102
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 10.1

01

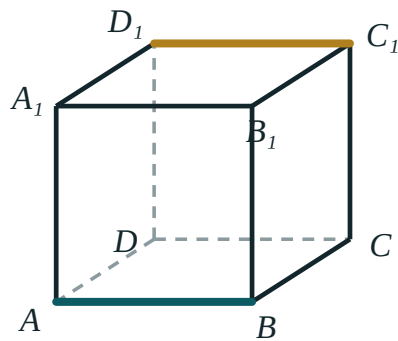
Explanation

The test

TEST FOR SKEW LINES

If one line lies in a plane and the other meets that plane at a point **not on** the first line, then the two lines are skew.

This is the practical test. Checking the definition directly — “no plane contains both” — requires ruling out infinitely many planes; the test requires exhibiting one.



AB and C_1D_1
never meet and
are not parallel:
they are skew

AB lies in the base; CC_1 meets the base at C, which is not on AB. Skew.

In cube $ABCD A_1 B_1 C_1 D_1$: AB lies in the plane $ABCD$, and CC_1 meets that plane at C , which is not on AB . So AB and CC_1 are skew.

The angle

Translate either line until it meets the other, and measure the angle between the resulting intersecting lines. By the theorem on respectively parallel sides, the answer does not depend on which line is moved or where it is moved to.

$$\text{angle}(a, b) = \text{angle}(a', b) \text{ whenever } a' \parallel a$$

In a cube, AB and B_1C_1 : translate B_1C_1 down to BC . Since $AB \perp BC$, the angle is 90° .
Two lines can be perpendicular without ever meeting.

TRANSLATE, DO NOT ROTATE

Only a translation preserves direction. Sliding a line along itself, or turning it, changes the answer.

The common perpendicular

THEOREM

Two skew lines have exactly one **common perpendicular** — a segment meeting both and perpendicular to both. Its length is the **distance** between the lines, and it is the shortest segment joining a point of one to a point of the other.

In a cube of side a , the common perpendicular of AB and CC_1 is BC , of length a . It meets AB at B and CC_1 at C , and is perpendicular to both.

A second method. Take the plane through one line parallel to the other. The distance between the skew lines equals the distance from the second line to that plane — a construction that generalises to cases where the common perpendicular is hard to see.

Distances in a cuboid

PAIR IN CUBOID $A \times B \times C$	COMMON PERPENDICULAR	DISTANCE
AB and CC_1	BC	b
AB and DD_1	AD	b
AA_1 and BC	AB	a
AB and D_1C_1	they are parallel	—

In a cube every edge has four skew partners, and in each case the common perpendicular is another edge. The picture is easier than the algebra, which is why the diagram is drawn first.

02 Worked examples

EXAMPLE 1 In cube $ABCD A_1 B_1 C_1 D_1$ of side 6, find the distance between AB and CC_1 .

1

BC meets AB at B and CC_1 at C .

2

$BC \perp AB$ and $BC \perp CC_1$.
It is the common perpendicular.

3

$|BC| = 6$

ANSWER

6

EXAMPLE 2 Are AD and B_1C_1 skew in a cube?

- ① $B_1C_1 \parallel BC$
Opposite sides of face BCC_1B_1 .
- ② $BC \parallel AD$
Opposite sides of the base.
- ③ By transitivity $B_1C_1 \parallel AD$.
- ④ Parallel lines are coplanar, so they are not skew.
Always test for parallelism first.

ANSWER

No — they are parallel

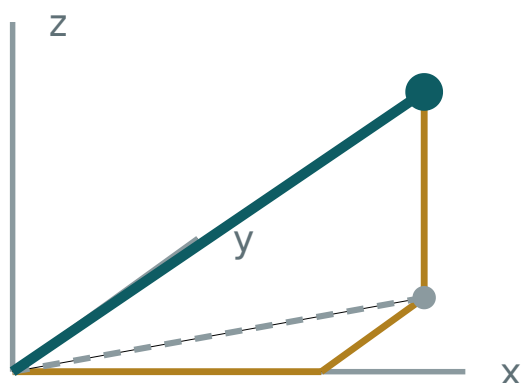
EXAMPLE 3 Cuboid $4 \times 6 \times 8$. Find the angle between AB and DB_1 .

- ① $AB = 4$ along x ; DB_1 from $D(0,6,0)$ to $B_1(4,0,8)$.
- ② Translate AB to DC , which meets DB_1 at D .
- ③ $DC = 4$, $DB_1 = \sqrt{16 + 36 + 64} = \sqrt{116}$
- ④ $\cos \theta = \frac{4}{\sqrt{116}} \approx 0.3714$

ANSWER $\approx 68.2^\circ$ **03 Interactive model**

Hold two pencils skew, then slide one parallel until they cross — the angle is unchanged.

Skew edges of a box



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to'g'ri chiziqlar	Скрещивающиеся прямые
Test for skew lines	Ayqashlik alomati	Признак скрещивающихся прямых
Common perpendicular	Umumiy perpendikulyar	Общий перпендикуляр
Distance between skew lines	Ayqash chiziqlar orasidagi masofa	Расстояние между скрещивающимися прямыми
Angle between skew lines	Ayqash chiziqlar burchagi	Угол между скрещивающимися прямыми
Translation of a line	Chiziqni ko'chirish	Перенос прямой
Shortest segment	Eng qisqa kesma	Кратчайший отрезок
Non-coplanar	Bir tekislikda yotmaydigan	Не лежащие в одной плоскости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The test for skew lines needs:

two planes

one line in a plane and the other meeting it off that line

a common point

equal lengths

Two skew lines have how many common perpendiculars?

0

1

2

infinitely many

The distance between skew lines is:

zero

the common perpendicular's length

infinite

undefined

To find the angle between skew lines you:

rotate one

translate one

project both

measure directly

Skew lines can be perpendicular:

never

always

sometimes

only in a cube

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the test for skew lines

ANSWER One line in a plane, the other meeting it off that line

2. Are AB and CC_1 skew in a cube?

ANSWER yes

3. Are AB and D_1C_1 skew?

ANSWER no — parallel

4. Common perpendicular of AB and CC_1

ANSWER BC

5. Distance between them in a cube of side 5

ANSWER 5

6. Angle between AB and B_1C_1

ANSWER 90°

7. How many edges are skew to a given edge of a cube?

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Cube of side 6: distance between AB and CC_1

ANSWER 6

2. Cube of side 6: distance between AA_1 and BC

ANSWER 6

3. Angle between AB and DC_1 in a cube

ANSWER 45°

4. Angle between AB_1 and BC_1 in a cube

ANSWER 60°

5. Cuboid $4 \times 6 \times 8$: angle between AB and DB_1

ANSWER $\approx 68.2^\circ$

6. Common perpendicular of AA_1 and CD in a cube

ANSWER AD

7. List the four edges skew to AD

ANSWER $BB_1, CC_1, A_1B_1, D_1C_1$

Hard · stretch and reason

7 PROBLEMS

1. Cube of side a : distance between AB and B_1D_1

ANSWER $\frac{a}{\sqrt{2}} \approx 0.707a$

2. Cube of side a : distance between the diagonals AC and B_1D_1

ANSWER a

3. Cube of side a : distance between AB and A_1C_1

ANSWER a

4. Prove the common perpendicular is unique

ANSWER Two would force the lines to be coplanar

5. Prove the common perpendicular is the shortest joining segment

ANSWER Any other joins via a right triangle with it as a leg

6. Cuboid $3 \times 4 \times 12$: distance between AB and CC_1

ANSWER 4

7. Explain the plane method for the distance between skew lines

ANSWER Take the plane through one parallel to the other; the distance is from the second to it

07 Homework

Homework — 5 tasks

Draw the cube once, large, and answer tasks 1–3 from it.

1. In cube $ABCD A_1 B_1 C_1 D_1$, list every edge skew to CD .
2. Find the common perpendicular and the distance between AD and CC_1 for side 8.
3. Find the angle between AD and BC_1 .
4. Cuboid $5 \times 12 \times 9$: find the angle between AB and DB_1 .

5. Explain in three sentences why the common perpendicular is the shortest segment joining the two lines.